Ziphozinhle Maqopi

400

100%

40

150%

Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ROTATING ELECTRICAL MACHINERY
SUMMATIVE ASSESMENT

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: Ziphozinhle Maopi	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Ziphozinhle Maopi

Learner's signature: Z Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

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Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Ziphozihle Maqopi	Signature
Assessors Name:	Signature:

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LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
1	a) List the different types of alternating current motors and describe their construction b) Describe with the aid of diagrams, the configuration of motor connections c) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	5 5 10

1a)

• Synchronous Motors: These motors have a stator with coils that generate a rotating magnetic field. The rotor is attached to the shaft and follows the magnetic field. synchronous motors are known for their high efficiency, ~~making~~ and ability to maintain a constant speed under varying loads

• Induction Motors: Induction motors ~~are known for their~~ have a stator with windings that create a rotating magnetic field. The rotor is attached to the shaft and is excited by the magnetic field.

• Brushless Motors: These Motors use a brushless system to reduce friction and noise. They have a stator with windings that create a rotating magnetic field and the rotor

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LEARNING UNIT 2:

KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)

Learning Outcome 1:	KT0201 Construction of direct current motors
Learning Outcome 2:	KT0202 Principles of operation for direct current motors
Learning Outcome 3:	KT0203 Configuration of direct current motor connections
Learning Outcome 4:	KT0204 Types and application of direct current motors
Learning Outcome 5:	KT0205 Testing principles of direct current motors
Learning Outcome 6:	KT0206 Protection of direct current motors
Learning Outcome 7:	KT0207 Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) Explain the principle of operation of direct current motors	2
	b) Describe and explain how various tests are conducted on direct current motors.	2
	c) Describe and explain protection devices used for direct current motors	3
	d) Compare the advantages and disadvantages of different types of direct current motors	3

2a) The principle : When a current-carrying conductor is placed in a magnetic field it experiences a force (Flemming's left hand Rule)

Operation : In a direct current motor, curren through the armature coil in the field produces torque. The commutator reverses current each half-turn to keep rotation continuous.

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b) No-load test: run the DC motor at rated voltage with no mechanical load to measure no-load current and speed. Used to find constant losses and Efficiency without loading the machine

Load test : Apply a mechanical load via a brake or dynamometer, measure input voltage or current and output torque

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LEARNING UNIT 3:

Rotating electrical machinery - Alternators and Generators (30%)

Learning Outcome 1:	KT0301 Construction of alternators and generators
Learning Outcome 2:	KT0302 Principle of operation of alternators and generators
Learning Outcome 3:	KT0303 Configuration of alternator and generator connections
Learning Outcome 4:	KT0304 Types and application of alternators and generators
Learning Outcome 5:	KT0305 Testing principles of alternators and generators
Learning Outcome 6:	KT0306 Protection of alternators and generators
Learning Outcome 7:	KT0307 Calculate properties of alternators and generators

Task	Questions Description	Marks
2	a) List the different types of alternators and generators and describe their construction	2
	b) Explain the fundamental differences between alternators and generators	2
	c) Describe and explain how tests are conducted on alternators and generators	3
	d) Describe with the aid of diagrams, the configuration of alternator and generator connections, power circuits and control circuits.	3

2a)

• Automotive alternators:

Construction: These are lightweight, designed to be mouted on an engine. They consist of a rotor (which generates a magnetic field) and a stator which contains the armature winding

• Brushless alternators:

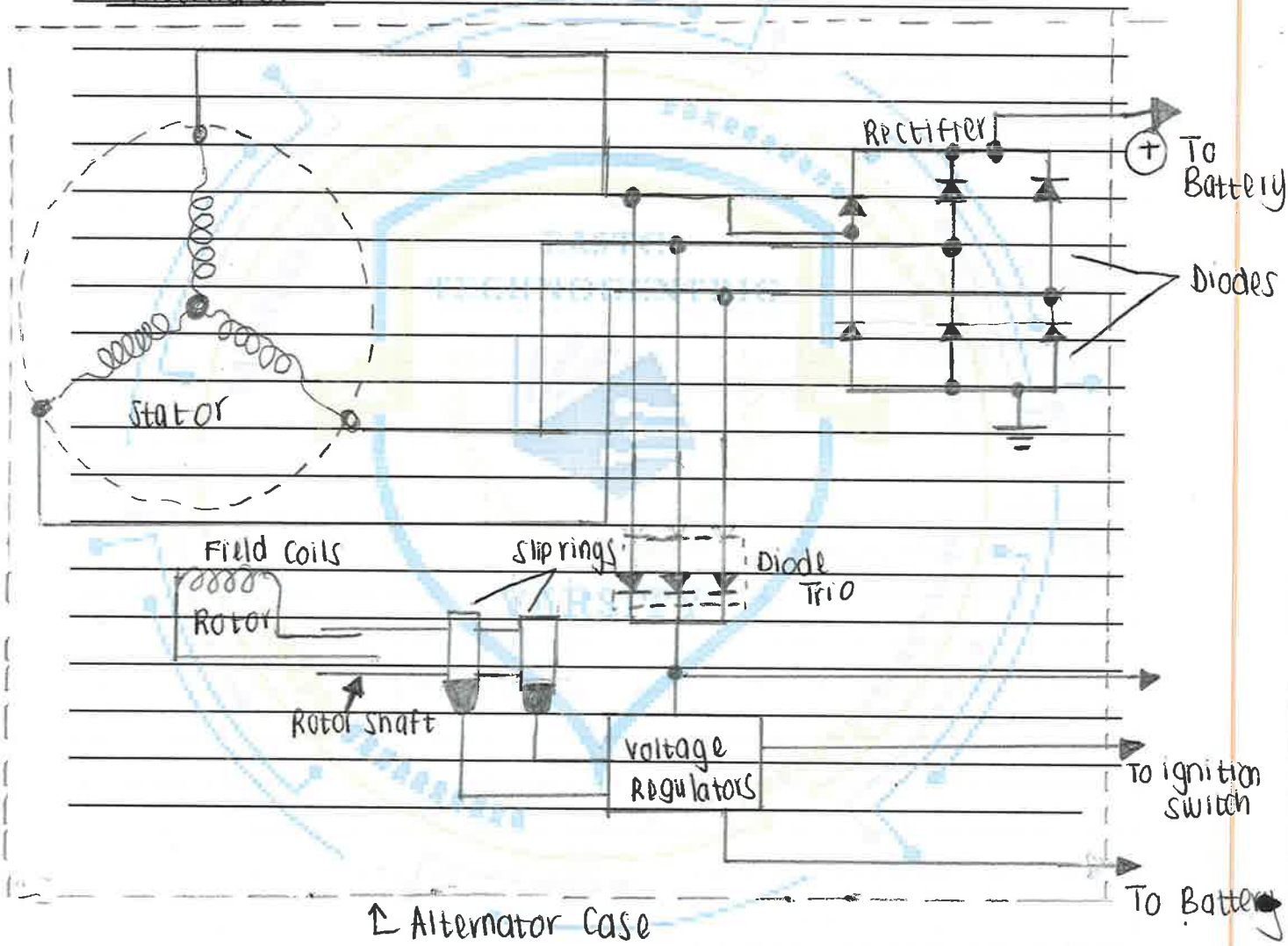
Construction: Brushless alternators remove the need for brusnes and commutators by using rotating magnetic field and a stationary armature, with the rotor typically made of permant

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3. Load test: The unit is connected to a variable load bank and operated from no-load bank and ~~—~~ to full-load. Voltage, frequency current, and temperature are ~~reco~~ recorded to verify stability, regulation and thermal performance

d) Alternator



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Candidate Feedback Report

Candidate's Name	Ziphozihle Maopi	ID No.	0802030118086
Assessor's Name		Reg. No.	

ASSESSMENT DECISION			
Source of Evidence	C	NYC	Comment
Knowledge questions	✓		Good
Product /Work sample	✓		Good
Project	✓		Good
Indirect Evidence	✓		Good
Overall Assessment Decision			
Additional Notes			
Date			
Signature of Assessor		Signature of Candidate	

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Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Ziphozihle Maqopi	ID No.	0802030118086

MODERATION DECISION

	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

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